

Compact space.

1. a). Let  $\mathcal{C}, \mathcal{C}'$  be topologies on  $X$ , with  $\mathcal{C} \subseteq \mathcal{C}'$ .

If  $(X, \mathcal{C}')$  is compact, then  $(X, \mathcal{C})$  is compact.

Proof. Let  $\mathcal{O} \in \mathcal{C}$  be an open cover of  $X$ . Then, since  $\mathcal{O} \subseteq \mathcal{C}'$  and  $(X, \mathcal{C}')$  compact,  $\mathcal{O}$  has finite subcover.  $\square$

1. b). If  $X$  is Compact  $T_2$  under both  $\mathcal{C}$  and  $\mathcal{C}'$ ,

then either  $\mathcal{C}$  and  $\mathcal{C}'$  are equal or they are Not comparable.

Proof. If  $\mathcal{C} = \mathcal{C}'$ , then there is nothing to prove. Suppose  $\mathcal{C} \neq \mathcal{C}'$ .

Enough to show that neither  $\mathcal{C} \subseteq \mathcal{C}'$  nor  $\mathcal{C}' \subseteq \mathcal{C}$ .

Suppose that  $\mathcal{C} \subseteq \mathcal{C}'$ . Then, we can choose an open  $U \in \mathcal{C}' \setminus \mathcal{C}$ .

Now,  $X \setminus U$  is closed in  $\mathcal{C}'$  but not  $\mathcal{C}$ . Since  $\mathcal{C}'$  is Compact  $T_2$ ,

$X \setminus U$  is Compact set of  $\mathcal{C}'$ . Since  $\mathcal{C} \subseteq \mathcal{C}'$ , for any open cover  $\mathcal{O} \subseteq \mathcal{C}$  for  $X \setminus U$ ,

it has finite subcover. Therefore,  $X \setminus U$  is compact in  $(X, \mathcal{C})$ , thus closed.

But, it is contradiction. Similarly,  $\mathcal{C}' \not\subseteq \mathcal{C}$ .

2. a). Every subspace in Co-finite Topology  $(\mathbb{R}, \mathcal{C}_f)$  is Compact.

b). In Co-Countable  $(\mathbb{R}, \mathcal{C}_c)$ , is  $[0, 1]$  Compact? an

Proof. Let  $A \subseteq \mathbb{R}$  be a subset. Let  $\mathcal{O} = \{\mathbb{R} \setminus F \mid F \text{ is finite}\}$  be open cover for  $A$ . There is,

$$A \subseteq \bigcup_{U \in \mathcal{O}} U \Leftrightarrow \bigcap_{U \in \mathcal{O}} \mathbb{R} \setminus U \subseteq A^c$$

But since  $\mathbb{R} \setminus U$  is finite for each  $U \in \mathcal{O}$ , there are finite number of  $U_i, \dots, U_n$  s.t.

$$\bigcap_{i=1}^n \mathbb{R} \setminus U_i \subseteq A^c \Leftrightarrow A \subseteq \bigcup_{i=1}^n U_i$$

Therefore  $A$  is Compact. (More specifically, for any distinct finite set  $F_1$  and  $F_2$ ,  $F_1 \cap F_2$  has number of less than  $\min\{\#(F_1), \#(F_2)\}$ .)

But, in the Co-Countable,  $[0, 1]$  is Not compact, because

Construct sequences; denote  $\mathcal{O} \cap [0, 1] = \{U_n \mid n \in \mathbb{N}\}$ , and let  $\mathcal{U}_n = (\mathbb{R} \setminus \mathbb{Q}) \cup \{x_n\}$

Then,  $\bigcup_{n=1}^{\infty} U_n \supseteq [0, 1]$  but it has no finite subcover.

3. Finite Union of Compact Subspace is Compact.

Proof. Let  $K_1, \dots, K_n \subseteq X$  be Compact subsets. Let  $\mathcal{O} \subseteq \mathcal{P}(X)$  be an open cover of  $\bigcup_{i=1}^n K_i$ .

Since  $\bigcup_{U \in \mathcal{O}} U \supseteq \bigcup_{i=1}^n K_i \supseteq K_j$  ( $1 \leq j \leq n$ ), for each  $j$ ,  $\mathcal{O}$  has a finite subcover for  $K_j$ ,

denote  $\mathcal{U}_j^1, \dots, \mathcal{U}_j^{r_j} \in \mathcal{O}$ . Now, clearly

$$\bigcup_{i=1}^n K_i \subseteq \bigcup_{j=1}^n \bigcup_{m=1}^{r_j} \mathcal{U}_j^m$$

4. Every Compact Subspace in a Metric space is closed bounded.

Give the example that the Converse fails.

Proof. Closed is given by: Compact in  $T_2$  is closed, because

Let  $X$  be a Hausdorff space, and  $K \subseteq X$  be a Compact subset.

Let  $x \in X \setminus K$  be fixed. Now, for each  $y \in K$ , there exists disjoint open  $\mathcal{U}_y^x, \mathcal{V}_y$  s.t.

$$x \in \mathcal{U}_y^x, y \in \mathcal{V}_y, \mathcal{U}_y^x \cap \mathcal{V}_y = \emptyset$$

Now,  $\{\mathcal{V}_y | y \in K\}$  is open cover of  $K$ , therefore there is a finite subcover

$$\{\mathcal{V}_{y_i} | i=1, 2, \dots, r_x\} \text{ and open neighborhoods of } x \{\mathcal{U}_{y_i}^x | i=1, \dots, r_x\}.$$

Put  $V_x = \bigcap_{i=1}^{r_x} \mathcal{U}_{y_i}^x$ . Then,  $V_x$  is open neighborhood and  $K \cap V_x = \emptyset$ , thus  $V_x \subseteq K^c$ .

$$\text{(More specifically, } K \subseteq \bigcup_{i=1}^{r_x} \mathcal{V}_{y_i} \text{ and } \left( \bigcup_{i=1}^{r_x} \mathcal{V}_{y_i} \right) \cap V_x = \bigcup_{i=1}^{r_x} (\mathcal{V}_{y_i} \cap V_x) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset.$$

And boundedness is clear.

Counterexample:

15. Let  $X$  be a Hausdorff, and  $A, B \subseteq X$  be disjoint compact subsets.

Then, there exist disjoint open sets  $U$  and  $V$  containing  $A$  and  $B$ , respectively.

Proof. Let  $x \in A$  be fixed. Then, for each  $y \in B$ , there exist opens  $U_{(x,y)}$  and  $V_{(x,y)}$  s.t.

$$x \in U_{(x,y)}, \quad y \in V_{(x,y)}, \quad U_{(x,y)} \cap V_{(x,y)} = \emptyset.$$

Since  $\{V_{(x,y)} \mid y \in B\}$  is open cover of  $B$ , there exists a finite subcover

$$\{V_{(x,y_i)} \mid i=1, 2, \dots, r_x\}. \text{ Now, } U_x = \bigcap_{i=1}^{r_x} U_{(x,y_i)} \text{ is open neighborhood of } x.$$

$$V_x = \bigcup_{i=1}^{r_x} V_{(x,y_i)} \text{ is open set containing } B, \text{ and } U_x \cap V_x = \emptyset. \text{ because}$$

$$U_x \cap V_x = U_x \cap \left( \bigcup_{i=1}^{r_x} V_{(x,y_i)} \right) = \bigcup_{i=1}^{r_x} (U_x \cap V_{(x,y_i)}) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset$$

In summary, for each  $x \in A$ , there are two disjoint opens  $U_x, V_x$  s.t.

$$x \in U_x, \quad B \subseteq V_x, \quad U_x \cap V_x = \emptyset.$$

And,  $\{U_x \mid x \in A\}$  is an open subcover of  $A$ , thus it has a finite subcover

$$\{U_{x_i} \mid i=1, \dots, m\}. \text{ Finally,}$$

$$\bigcup_{i=1}^m U_{x_i} \text{ and } \bigcap_{i=1}^m V_{x_i} \text{ are disjoint open sets containing } A \text{ and } B, \text{ respectively.}$$

$$\text{Disjointness is given by } \left( \bigcup_{i=1}^m U_{x_i} \right) \cap \left( \bigcap_{i=1}^m V_{x_i} \right) = \bigcup_{i=1}^m \left( U_{x_i} \cap \left( \bigcap_{i=1}^m V_{x_i} \right) \right) = \bigcup \emptyset = \emptyset.$$

16. If  $f: X \rightarrow Y$  is continuous map where  $X$  is compact and  $Y$  is Hausdorff, then  $f$  is closed map.

Proof. Let  $C \subseteq X$  be a closed set. Since  $X$  is compact,  $C$  is compact.

And, the continuous map  $f$  preserves compactness,  $f[C]$  is compact in  $Y$ , thus closed.

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